Jay Desai

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Summary

Physicist with **six years** of research experience in statistical modeling and large-scale computation (PhD expected December 2026), with **four** peer-reviewed journal publications, **70+** citations, collaborated in multi-institutional research teams and invited talks at Brookhaven National Lab and CERN. Experienced in building end-to-end computational pipelines, combining LLM tooling, sentence-transformer embeddings, and rigorous statistical evaluation.

Education

- **Stony Brook University** 2020 – Present
  Ph.D. in Physics (High Energy Theory and Phenomenology)
- **Stony Brook University** 2018 – 2020
  M.A. in Physics
- **Indian Institute of Technology Roorkee** 2014 – 2018
  B.Tech., Electronics and Communication Engineering, Minor in Physics

Research Experience

- **Graduate Research Assistant, C.N. Yang Institute for Theoretical Physics** 2020 – Present
  Stony Brook University
  - Developed an end-to-end analysis pipeline: derived model predictions symbolically in Mathematica, fed them into MadGraph for Monte Carlo event generation at scale (**1M+** events), and processed simulation output for statistical fits and visualization in Python and C++.
  - Aggregated and harmonized **85+** measurements from heterogeneous experimental sources into a unified statistical analysis, accounting for correlations among inputs.
  - Performed global likelihood fits over an **8**-dimensional parameter space via chi-squared minimization; profiled and marginalized nuisance directions, extracted confidence regions, and identified and resolved parameter degeneracies via reparametrization.
  - Constructed dimension-eight basis of **175** operators for universal effective field theories filtered from **44,807**. Cross-checked using group theory and representation theory techniques to compute group invariants.
- **Research Assistant, Center for Nuclear Science** 2018 – 2020
  Stony Brook University
  - Experience with large-scale simulation packages generating **1M+** events and C++ based data analysis framework ROOT, executed via HTCondor job submission on the Brookhaven National Lab computing cluster.
  - Applied statistical reweighting techniques to estimate an uncertainty reduction of **20**% for the proposed electron-ion collider.

Technical Skills

- **Machine Learning**: PyTorch, Scikit-learn, Matplotlib, Pandas, Jupyter, statistical inference, high-dimensional modeling, model validation, data analysis
- **Programming**: Python, C, C++, Java, SQL
- **Mathematical Methods**: Probability theory, stochastic processes, hypothesis testing, multi-parameter regression
- **Scientific Computing**: SciPy, NumPy, Mathematica, MATLAB, ROOT, IBM Qiskit
- **High-Performance Computing**: HTCondor, Linux cluster computing
- **Simulation & Modeling**: Monte Carlo Methods: MadGraph
- **Tools**: Git, LaTeX, Claude Code

Conference/Invited Talks

- **Brookhaven National Lab (BNL) Seminar (invited), Brookhaven, December 11th 2025**
  "Universal Standard Model Effective Field Theory beyond dimension-six"
- **Higgs and Effective Field Theory (HEFT), CERN, June 3rd 2025**
  "Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory"
- **Large Hadron Collider (LHC) EFT Working Group meeting, (invited), April 28th 2025**
  "Drell-Yan production in universal theories beyond dimension-six SMEFT"
- **American Physical Society (APS) DPF-Pheno 2024, University of Pittsburgh, May 13th 2024**
  "Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory"
- **American Physical Society (APS) April Meeting, 2020**
  "Strange quark polarized distribution function using SIDIS at EIC"

Projects

- **Predicting Award Nominations for Sci-Fi/Fantasy Novels** Erdos Institute Data Science Boot Camp
  GitHub
  - Built learning-to-rank models (LightGBM/XGBoost LambdaRank, grouped by year) in a team of five to predict Hugo and Locus Award finalists from 50+ years of book data with severe class imbalance (1.3% positives), achieving a 1.7× improvement in F_1 over logistic regression and prior-nominee baselines.
  - Engineered features from multi-source data (ISFDB SQL dump, Hardcover API, SFADB): year-cohort z-scored author/publisher prestige metrics, cosine similarity to past nominees in tag space, and 384-dim synopsis embeddings (all-MiniLM-L6-v2 sentence transformer), with custom sklearn transformers to prevent temporal leakage.
  - Evaluated via nested walk-forward time-series cross-validation (train 1971–2014, test 2015–2025) using ranking metrics (Precision@k, NDCG@k) alongside F_1 score; diagnosed and fixed a data leakage bug in cumulative author-history features.
- **Forecasting 45-Day Forward Stock Beta**
  Quant assessment mini-project
  - Built an end-to-end ML pipeline to predict 45-day forward-looking beta relative to SPY from daily price/volume panel data; engineered 15 features across multi-horizon trailing betas, correlation/volatility decomposition, beta instability, market-state, and liquidity/momentum groups, with universe cleaning to remove leveraged/inverse ETFs contaminating the loss.
  - Combined ridge regression and LightGBM into a 50/50 ensemble evaluated via purged walk-forward cross-validation (45-day purge to prevent label-overlap leakage); on the held-out test period the ensemble reduced winsorized MSE by 2.6% over a fitted Blume-shrinkage baseline.
- **Multilayer Perceptron from Scratch** CSE 353 Machine Learning
  Stony Brook University
  - Implemented a multilayer perceptron in PyTorch with manual forward/backward passes and gradient derivations; trained and validated on classification datasets, evaluating generalization across decision boundaries.

Publications

All papers can be found on my INSPIRE and Google Scholar profile. Authors ordered alphabetically.

- *"USMEFT as a tool for discovery of universal new physics at high luminosity LHC"*
  arXiv:2605.15262 [hep-ph], Phys. Rev. D 114, 055055 (2026)
  Tyler Corbett, **Jay Desai**, O. J. P. Eboli, M. C. Gonzalez-Garcia, Matheus Martines
- *"Drell-Yan production in universal theories beyond dimension-six SMEFT"*
  arXiv:2503.19962 [hep-ph], Phys. Rev. D 112, 013009 (2025)
  Tyler Corbett, **Jay Desai**, O. J. P. Eboli, M. C. Gonzalez-Garcia, Matheus Martines, Peter Reimitz
- *"Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory"*
  arXiv:2404.03720 [hep-ph], Phys. Rev. D 110 (2024) 033003
  Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia
- *"Impact of dimension-eight SMEFT operators in the EWPO and Triple Gauge Couplings analysis in Universal SMEFT"*
  arXiv:2304.03305 [hep-ph], Phys. Rev. D 107, 115013 (2023)
  Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia, Matheus Martines, Peter Reimitz

Awards & Achievements

- Gerald Brown Prize (Research Travel Award), Stony Brook University
- Nationwide Education and Scholarship Test (Ranked **13**th nationally)
- Joint Entrance Examination (JEE) Advanced (Ranked **971** nationally)

Leadership & Teaching

- Instructor for PHY 121, PHY 122, PHY 133, PHY 134.
- Teaching Assistant for PHY 301, PHY 308, PHY 503.
- Journal Referee for the Journal of High Energy Physics.